Final Report

Integrative Project in Aerospace Engineering II

Bachelor's in Aerospace Engineering

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Abstract

This semester, within the context of the Integrative Project Curricular Unit of the second year of the Aerospace Engineering Degree, the team was challenged to make a rocket from scratch that fulfilled all the requirements imposed by Project Coordinators. First, it was essential to build a group identity to strengthen interpersonal relationships within the team. An acronym was chosen as the team's designation, MARS – Minho Aerospace Rocketry Systems, reflecting our identity as both UMinho students and engineers in our own "company". Then followed the logo which represents our team's name not in an obvious way but invoking the power of a deity, Mars, the Roman god of war. In this project, we developed our research and information gathering skills and acquired notions about engineering projects. Based on our continuous work of scientific research, various designs, and numerous simulations, in the end, the team managed to design, test through simulations, build and reproduce a final model of a rocket that hopefully will take off, fly between the given threshold, and land safely, thus achieving the goal that we set for ourselves.

1. Introduction

The Mars (Minho Aerospace Rocketry Systems) team, consisting of students Beatriz Carvalheira, Filipe Ferreira, Francisco Araújo, Jorge Azevedo, Kseniia Kontseva and Pedro Meleiro, undertook a project at the University of Minho as part of the Integrated Aerospace Engineering Project course. The goal of this project was to build a rocket from scratch that fulfilled all the requirements imposed by the teachers. To briefly explain the object of development, a rocket is a specialized system designed to propel payloads, such as satellites or spacecrafts into space by means of rocket propulsion. However, we will be assembling a much smaller version of this.

Each second semester of the Aerospace Engineering degree includes an Integrative Project, which follows the Project-Based Learning (PBL) approach. This project aims to help students assimilate essential aerospace knowledge and develop a broad understanding of the field. It evaluates skills such as autonomy, teamwork, proactivity, critical thinking, and leadership. It also provides practical project experience, allowing students to apply theoretical knowledge from other courses to build, manage, and develop a new model. Additionally, it helps students get closer to professional work and research, encourages them to explore beyond the curriculum, and improves their teamwork and interpersonal relationships.

At the beginning of the project, we were given some requirements we had to follow meticulously: the rocket had to achieve a minimum target altitude of 150 meters and not exceed an altitude higher than 250 meters; it couldn't exceed a 2 meter length; it also needed to safely launch and land a payload of 100g and, finally, it had to be recovered unbroken. The rockets will then be launched in the PAM-AERO activity which will occur in Aerodrome of Cerval.

Moreover, given the importance of sustainability in science and engineering, we managed our project with the aim of adhering to the 17 Sustainable Development Goals set by the United Nations General Assembly as much as possible.

2. Altimeter

In the world of model rockets, barometric altimeters stand out as essential instruments for collecting altitude data during flight. By accurately measuring atmospheric pressure, which decreases with altitude, these altimeters provide valuable information for:

- **Evaluate rocket performance:** By comparing the actual altitude with the estimated altitude, you can identify the efficiency of your design and areas for improvement.
- **Analyse the flight trajectory:** Altitude data, combined with other sensors, make it possible to reconstruct the rocket's trajectory, from launch to landing.
- **Meet competition requirements:** In model rocketry events, the use of barometric altimeters may be mandatory to prove compliance with specific altitude objectives.

The operating principle of barometric altimeters is based on **Boyle's Law**, which relates the pressure and volume of a gas in a closed system at constant temperature. As the rocket rises, atmospheric pressure decreases, causing the gas inside the altimeter to expand. This expansion is detected by an internal sensor, which converts the information into an electrical signal. This signal, in turn, is processed by a microcontroller, which calculates the altitude based on the measured atmospheric pressure and the previously configured reference pressure.

Barometric altimeters for model rockets offer several advantages:

- **Simplicity:** Operation based on a simple physical principle, making them easy to understand and use.

- **Low cost:** Generally, more affordable than other types of altimeters, such as GPS.
- **Lightness and compact size:** Ideal for integration into rockets with limited space.
- **Low energy consumption:** Allows long operating times with a single battery.

On the other hand, it is important to consider some disadvantages of barometric altimeters:

- **Limited Accuracy:** Accuracy may be affected by variations in temperature and atmospheric conditions.
- **Simplified atmospheric models:** The formula used to calculate altitude assumes an idealized atmospheric model, which can lead to small inaccuracies at high altitudes.
- **Necessary calibration:** It is crucial to calibrate the altimeter periodically to ensure accurate measurements.

This project aims to construct an altimeter using the Raspberry Pi Pico microcontroller. The system is designed to measure altitude through a BMP280 sensor, record the collected data onto an SD card, and play a melody upon completing the sampling process. The code developed for this project includes several essential libraries: "RPi_Pico_TimerInterrupt.h" for handling timer interrupts, "Adafruit_Sensor.h" and "Adafruit_BMP280.h" for interacting with the BMP280 sensor, "SPI.h" and "SD.h" for communication and control of the SD card, and "pitches.h" for defining musical notes for the buzzer.

Initially, time intervals are defined for various activities, such as a sampling interval of one minute, a sampling period of 15 milliseconds, and a LED/buzzer period of 500 milliseconds. A 30-second delay is configured before starting the sampling, and the buzzer frequency is set to 1000 Hz. For altitude calculation, the sea-level pressure is set to 1010 hPa (This base pressure will need to be rectified on the day of the flight). Additionally, pins for the buzzer and the SPI interface for the SD card are configured.

The code uses several global variables, including instances for the BMP280 sensor, a log file for the SD card, and timers for the LED/buzzer and sampling. Flags are used to control the state of the buzzer and the sampling process. Specific functions are defined to initialize the system components, such as UART, GPIO, timers, SD card, and BMP280 sensor, as well as a function to play a melody.

In the **setup()** function, the necessary components are initialized. The UART is configured for serial communication if debugging is enabled. The pins for the LED and buzzer are set up, and

timers are adjusted to control the LED/buzzer and sampling. SPI communication with the SD card is established, and the log file is opened. The BMP280 sensor is initialized, and its sampling settings are configured.

The main function **loop()** controls the program's flow. Before launch, the system waits for an initial 30-second interval. During sampling, the buzzer is activated, and the sampling flag is set. The sampling timer is set to collect data at regular 15-millisecond intervals. At each interval, temperature and altitude data from the BMP280 sensor are read and recorded onto the SD card. After sampling, the timers and buzzer are deactivated, the log file is closed, and a melody is played to indicate the end of the sampling process.

The timer handlers are responsible for toggling the LED and controlling the buzzer during the sampling process. The sampling handler sets the **take_sample_now** flag, indicating that it is time to take a new sample. The initialization functions set up serial communication, GPIO pins, timers, SPI communication with the SD card, and the BMP280 sensor, and write the initial data to the log file.

The **play_tune()** function defines a melody with musical notes and their durations, using the buzzer to play the melody at the end of the sampling process.

In summary, the code developed allows for the construction of a functional altimeter using the Raspberry Pi Pico and the BMP280 sensor. The system collects and records altitude data at regular intervals, ensuring the necessary accuracy for applications in rocket projects or other activities involving altitude measurement. Additionally, the functionality of playing a melody at the end of the sampling process provides clear auditory feedback that the procedure has been successfully completed.

Through detailed analysis, we uncover the secrets behind collecting and recording altitude data during the flight of a model rocket.

This project stands out for being an excellent introduction to the world of microcontrollers, offering a rich learning experience and opening doors for the exploration of several innovative projects.

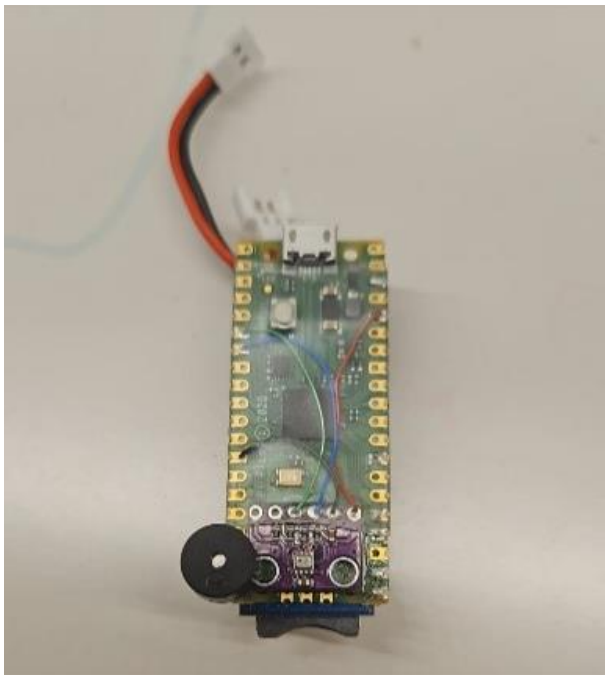

Figure 1 - Image of the altimeter

3. Construction Plan

Nose Cone and Payload Bay

The nose cone, payload bay and pins were 3D printed as planned, following the design choice in our last report. We ordered the parts from the university's multimedia services.

The nose cone was sanded thoroughly. This was made to remove the small imperfections left by the 3D printing process, which would've otherwise significantly increased the skin friction drag on this component. The shoulder part which connects to the payload bay from the outside and to the body tube from the inside was also sanded, along with the four pins, so that all parts fit nicely together – extra caution was taken on the outside of the shoulder, as well as the top surface of the pins, because the fit with the body tube cannot be too tight, otherwise it may lead to a catastrophic failure during the ejection of the parachute (this was also done while keeping in mind that, before the flight, a small piece of tape will be added on the shoulder in order to secure the pins).

When the nose cone assembly is ejected along with the parachute, it'll travel some distance where it can dissipate some energy and slow down (because of air resistance, friction with the body tube and shock cord elasticity), but, more likely than not, it'll still be travelling relatively fast by the

time the shock cord is fully taut. At that point, the assembly must resist a snap and a quick deceleration, which will result in a force that pulls the nose cone apart from the payload bay. The pins serve to ensure that these two pieces remain firmly attached, and so they can't break under that force. They were printed in a direction such that the different layers were perpendicular to the main forces acting upon them.

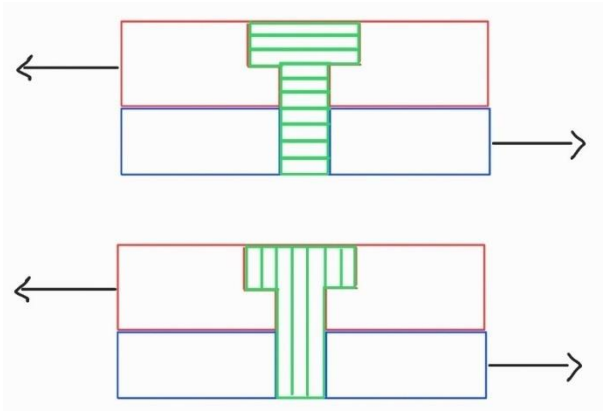

Figure 2 – Weak layer direction (on top) vs strong layer direction (on the bottom)

This is because the weakest direction of a 3D printed material is that which is parallel to its layers. It is intuitively easier to separate two layers than it is to break all of them. Thus, we concluded that we should print our pins like the one on the bottom of the figure, instead of the one on the top. Furthermore, thanks to our materials class from this semester, we know that the forces acting on the pin will result in a shear stress. The shear stress is given by the expression: $\tau = \mathrm{FA}$, where F is the force pulling the assembly apart and A is the area of the pins that opposes this pull. We can rewrite the expression as $\tau = \frac{\mathrm{F}}{N.\pi.r^2}$, where N is the number of pins used and r is the pin radius. We consulted the shear strength of PLA prints on a study published in the Journal of Materials Research and Technology, by Qinglin Wang et al., and the value published was 25.74 ± 1.03 MPa. Solving the equation for F, using the minimum value for shear stress, four pins, and our pin radius of 4 mm, we get a theoretical maximum pull force of 1242 N, which is a lot more than what is expected. Of course, we can't trust our pins to withstand forces around this value, as there are some factors that this calculation doesn't consider, like the possibility for printing errors or an uneven load distribution between different pins, and that is why we chose a pin number and radius that provide such a large safety margin.

We realized after printing that it would've been possible to design our nose cone and payload bay with screw threads instead of pin holes. We had chosen not to do this because we were advised by the university personnel that the printing precision might not be enough, so we went with the safest option we had. However, in the future we would like to try out this option, as it makes for a simpler design, with less parts and less possibilities for failure (the pins come with small risks like getting lost, coming off during flight, taking too long to assemble after turning on the altimeter, or even protruding out of the shoulder enough to somehow stop it from ejecting). That being said, we took every precaution in order to mitigate said risks, and we're confident that this system will work properly towards achieving our requirements.

Figure 2 - Image of the nose cone

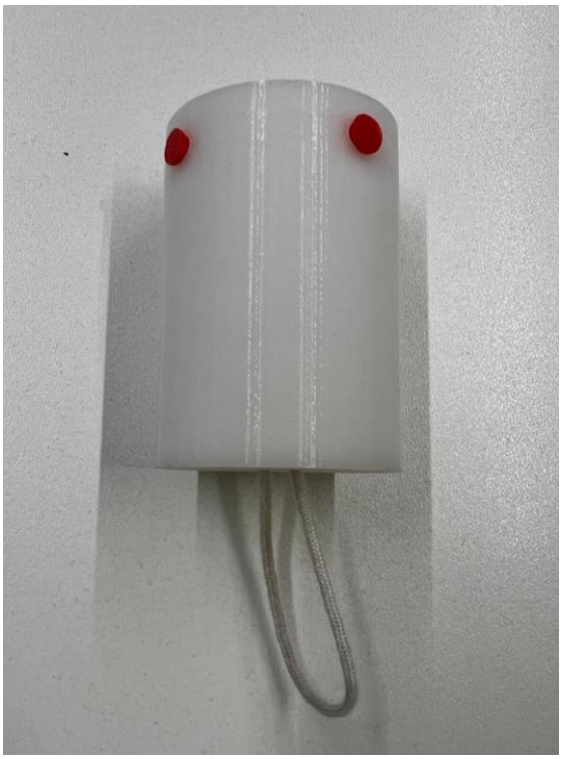

Figure 4 - Image of the payload bay

The payload bay features two small holes on the bottom, through which we tied a segment we cut out of our shock cord. This loop is used to tie the parachute and the shock cord.

Finally, we need to add weight on the tip of the nose cone, to limit the apogee altitude and increase the passive stability. To do this, we added a 75 g mixture of epoxy resin and steel balls, which we acquired from old ball bearings.

Motor Mount

The inner tube was cut using the same handsaw used for the body tube. It was also thoroughly sanded in the areas where we were going to glue other parts, ensuring a stronger bond.

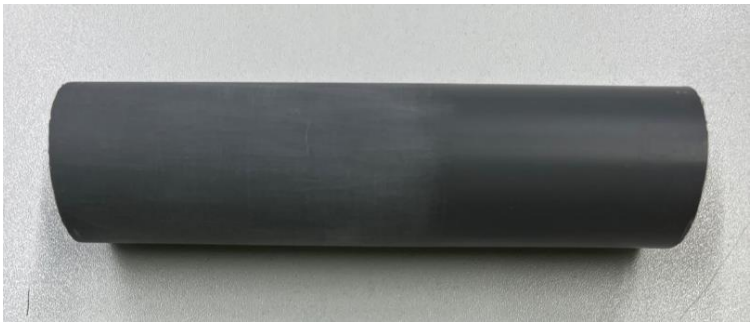

Figure 5 - Image of the inner tube

The motor block stops the motor from sliding up the inner tube and into the rocket. It needs to be reasonably strong, as this is the direction the motor will want to go during the flight. It also needs to be fire resistant, as it is the first point of contact with the ejection charge.

To make our motor block, we employed a technique which had already been previously used by a group member, during last year's project. The technique consists of rolling paper around a desired shape while applying a small amount of white glue all throughout. Once the glue dries out, you are left with a hardened hollow cylinder which you were able to make with high precision. We tested this part for fire resistance, and it held up exactly like we wanted.

Figure 6 - Image of the motor block

To build our centering rings we started by drawing two circles (for each centering ring) on the balsa board, we then proceeded to cut outside the largest circle with a small margin, and finally sanded until we achieved a satisfactory circular shape. For the inner circle we employed the same techniques of cutting and sanding, but instead of cutting outside the circle we cut inside and sanded outwards to achieve the desired shape. Both rings were sanded together, as to ensure that the motor faces directly down. After all that we applied a paper skin to the centering rings. We found this to be a great strategy to increase the strength of our rings without adding hardly any weight.

Figure 7 - Image of the centring rings

Our motor hook was made for a piece of metal provided by the Rocketeers group. We cut it and shaped it according to our design.

Figure 8 - Image of the motor hook

All parts were glued together using epoxy, which was given the necessary time to cure.

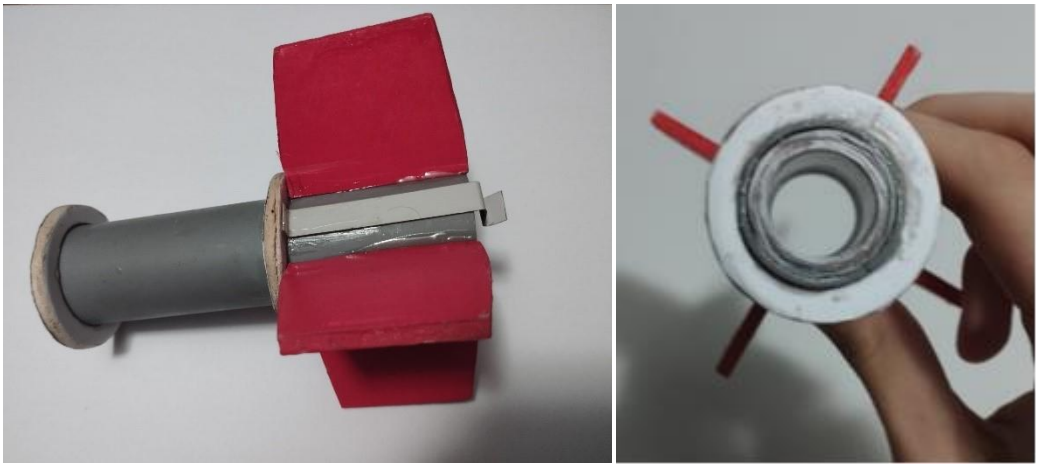

Figure 9 & 10 - Image of the motor mount

Fins

The fins were made exactly like we had planned in our preliminary design. We started off by drawing the shapes on the balsa sheet, then cut them with a small margin, and then sanded all the fins together so that they would be identical in shape. Next, we followed the same drawing and cutting procedure on a standard sheet of paper and applied the skins using wood glue we had bought for last year's project. We then had to sand the paper edges off and spray paint the four fins.

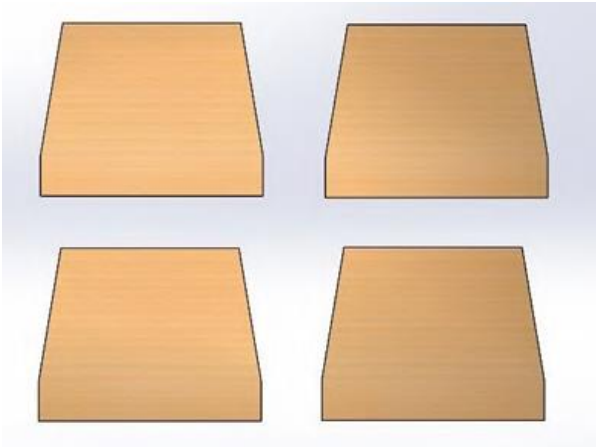

Figure 11 - Image of the fins made from balsa

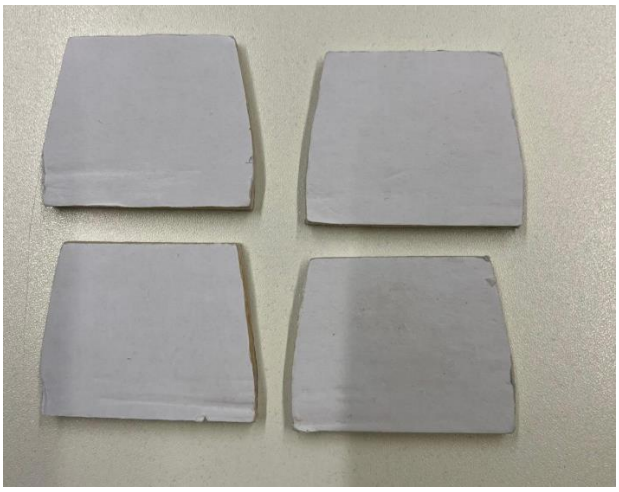

Figure 12 - Image of the fins with the paper skin

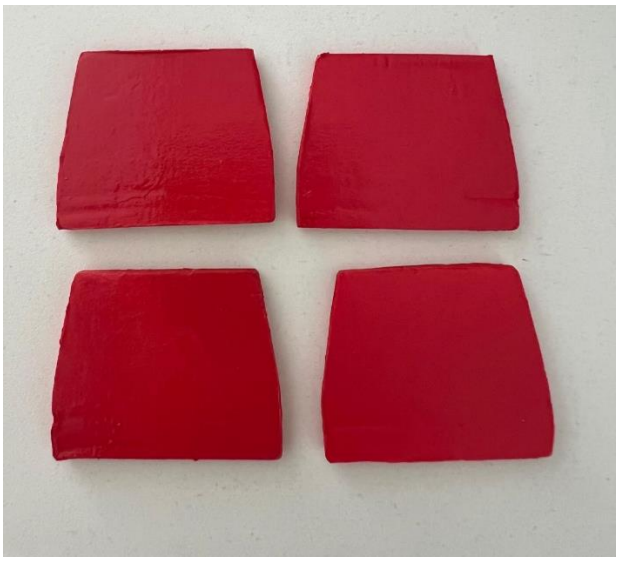

Figure 13 - Image of the fins already painted

In the meantime, we also had to cut four slits in the body tube for our fins to fit through. This was one of the most challenging parts of the build, as it requires better than millimetric precision – on this note, because our body tube has a 50 mm diameter meaning a perimeter of approximately 157 mm, an error of a single millimetre results in an angular deviation of over 2°, which can greatly impact roll, pitch, and yaw stability.

We calculated the tube's perimeter and marked it out on a sheet of paper. Then, we divided this distance by four, marking out the position for each fin. 2 mm were also drawn on either side of these initial marks, centring them on the 4 mm fins. Finally, we wrapped this paper around our tube and used it to mark where to cut the slits.

Figure 14 & 15 – On the left side an image of the body tube and on the right the method we used to mark the places to cut the slits

After ensuring that the marks were properly placed, we cut the tube using a handsaw blade. This process was very tedious, and it required a lot of concentration and patience, as it was very easy to deviate from the guidelines, and a tiny error could ruin the body tube. Nevertheless, we managed to cut all our slits within an acceptable margin for error.

Finally, we had to glue the fins to the motor mount while, once more, ensuring proper alignment. To do this, we cut a mould from our balsa board, which we used to keep the fins fixed in the correct position while the gluing agent cured. To glue them to the inner tube, we used epoxy resin, which allowed for a very strong bond.

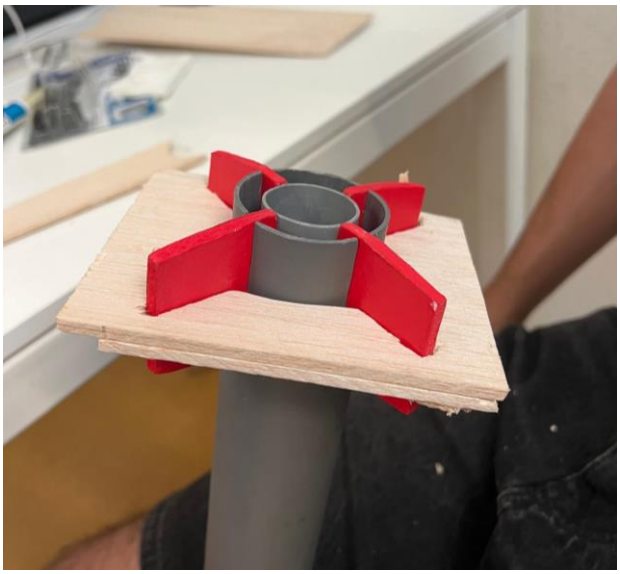

Figure 16 - Image of the mould we used to keep the fins fixed in the right position

Recovery System:

Parachute

For the parachute we had to options, either making it in-house or buying it from a shop. From the beginning, our priority was to cut costs as much as possible by doing as much as we could ourselves, ensuring that we only spent on what was necessary. Upon conducting a quick search online, we found that most of the parachutes available, although being affordable, were available in the USA. This meant we would have to import them, which would raise the overall cost due to shipping and taxes. Consequently, manufacturing the parachute ourselves emerged as not only the cheaper option but also an opportunity for us to gain valuable know-how on this component, which could be beneficial for future projects.

After purchasing all the materials necessary to fabricate the parachute, which included nylon fabric of red colour, duct tape and cotton thread and gathering all the tools to help the assembly, we started the process.

As we wanted an eight-sided parachute, we started with drawing an octagonal shape on the nylon fabric with a diameter of 78.2 cm and proceeded to cut it.

Figure 17 – Cutting the fabric in an octagonal shape

After that we moved on to the next step which entailed reinforcing the corners of the parachute with duct tape to ensure that the hole to put the shroud lines through, would not rip open during flight, using as much little tape as necessary as we could so we wouldn't add unnecessary weight to the rocket.

Figure 18 – Reinforcing the corners of the parachute

Then we needed to make the holes where the shroud lines would attach and for that we used an old nail that we heated up.

Afterwards, we began to cut the shroud lines. We wanted them to have the same length of the diameter of the parachute, but because we had to tied them to thread of the payload bay and

to the shock cord and we also would have to burn their ends so that the thread would not fray, we had to cut them a little longer, so we measured them to have 85 cm.

Finally, we put in the shroud lines in each hole, made a knot and burned the end of the knot to make sure that the knot wouldn't come undone during flight.

Figure 19 – Burning the end of the knots

Before attaching the shroud lines to the shock cord, we proceeded with the drop-tests, that followed the protocol already explained in the CDR, which entailed attaching the parachute to a dummy mass of similar weight and configuration of the rocket, in our case it had 530 g, which is the estimated weight at landing (with motor, but without fuel). In this case we used a water bottle with water in it and we launched it from a height of six storeys, which was the tallest building we could find in the campus.

Figure 20 & 21 – Drop-tests

To understand the landing dynamics and the accuracy of the data, we analysed a video where two frames are 20 frames apart, which corresponds to 2/3 of a second. The estimated drop height was between 2 and 3 meters. Using this information, we calculated a ground-hit velocity between 3 and 4.5 m/s.

To perform the video analysis and extract the necessary data, we used VLC media player, a versatile software that allows frame-by-frame analysis, essential for accurately calculating the impact velocity.

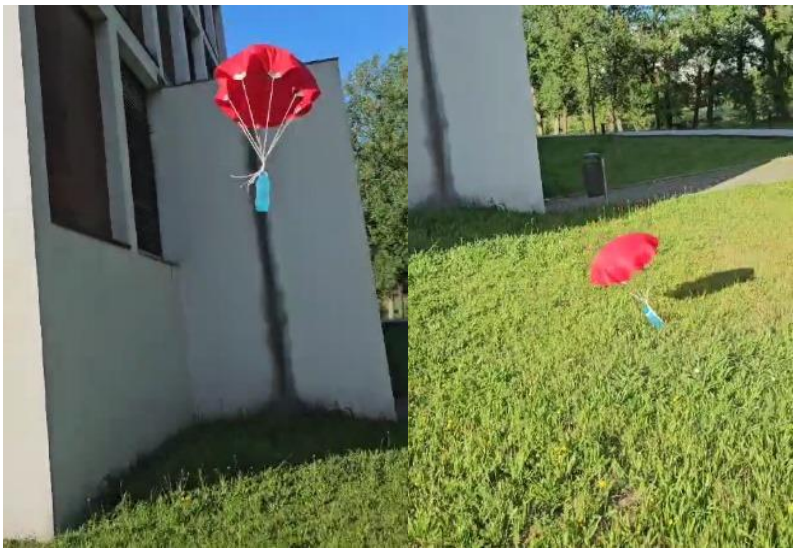

Figure 22 & 23 – Set of images used to estimate the ground-hit velocity

The drop-tests conducted on the parachute yielded satisfactory results, confirming its functionality and reliability. The parachute deployed correctly and demonstrated consisted performance across multiple trials, meeting the established safety and performance criteria. These outcomes validate the effectiveness of our design and manufacturing process.

Shock cord

In our pursuit of building all the rocket systems and keeping costs low, we decided to manufacture our own shock cord. However, none of us had the necessary expertise for this task. To overcome this challenge, we reached out to Professor Joaquim Jorge Gomes Peixoto from the Department of Textile Engineering at University of Minho.

With his expertise and guidance, we developed a shock cord composed of two segments. The first segment, measuring 1 meter in length, is made of aramid fiber. This choice was due to aramid's excellent fire resistance, making it ideal for the part closest to the rocket motor. The aramid fiber is braided with aramid fill to provide tensile strength capable of towing cars (as shown in the presentation). This ensures that the cord can withstand the high temperatures and mechanical stresses encountered during the rocket's operation.

The second segment is 60 centimetres long and also uses aramid as the core material. However, this segment is braided with elastic thread, allowing it to stretch and absorb shocks more

effectively. This design is crucial for the recovery phase of the rocket, where the shock cord must absorb the force of the parachute deployment without snapping or causing damage to the rocket.

This shock cord was manufactured using braiding machines and raw materials provided by the university. Through this project, we not only enhanced our product development skills but also expanded our networking capabilities, establishing valuable connections within the academic and engineering community. Collaborating with Professor Peixoto and leveraging the resources of University of Minho allowed us to gain insights into advanced textile engineering techniques and apply them to our aerospace project.

By developing the shock cord in-house, we were able to tailor its specifications precisely to our needs, ensuring optimal performance and reliability. This hands-on experience also deepened our understanding of material properties and their applications in high-stress environments.

Figure 24 & 25 - Image of the machine where the shock cord was made

Figure 26 – Process of braiding the Kevlar and aramid

Figure 27 - Shock cord

Figure 28 – Test of the resistance of the shock cord

4. Safety Flight Plan

This section of the report states the Rocket's Launch Safety Plan, which details the specific safety measures and risk mitigation strategies employed for the Rocket. Our goal is to ensure the safety of all personnel, equipment, and the surrounding environment through meticulous planning and the implementation of safety procedure.

In this Safety Plan, we will address the following key areas:

- Risk Assessment
- Mitigation Measures

This safety plan's risk assessment section identifies and assesses any possible dangers related to the Rocket launch. Identifying hazards, evaluating their likelihood and potential impact, and putting the right mitigation measures in place are the goals of this methodical approach. Through a comprehensive comprehension and management of these hazards, we can augment the security and triumph of the launch endeavour. This section will list the hazards that have been identified, assign probability based on past performance and professional analysis, and describe the tactics used to successfully reduce these risks. By taking this proactive stance, we make sure that all essential safety measures are done to safeguard people, property, and the surrounding area throughout the whole rocket launch process.

To achieve this, we have created a table that lists possible risks:

Pre-Launch	**Weather Conditions**	Weather conditions that could impact the launch or flight of the rocket
	Structural Integrity Compromise	Risk of structural damage or compromise to the rocket due to mishandling or transportation issues
	Disassembly of Components	Disassembly of critical rocket components
Launch	**Ignition System Malfunction**	Issues with the ignition system can prevent the rocket engines from starting
	Range Safety Issues	Failure to maintain proper clearance
	Launch Pad Malfunction:	Malfunction of launch pad systems
	Objects Striking Personnel	Objects becoming ejected from the rocket or launch pad infrastructure, potentially striking personnel
Post-Launch	**Unsuccessful Attempt In Reaching Target Altitude**	Failure in reaching 150-250 meters of altitude
	Altimeter or Data Storage System Failure	Loss of crucial flight data
	Landing at High Velocity	Failure to decelerate to a low velocity during landing can result in a hard landing

Table 1 – Possible risks during pre-launch, launch and post-launch

<u>Pre-Launch Phase:</u>

- **Weather Conditions**

Mitigation Measure: Real-time weather monitoring to track atmospheric conditions leading up to the launch. The launch will be postponed or rescheduled if adverse weather, such as high winds or heavy rain is forecasted.

- **Structural Integrity and Disassembly of Components**

Mitigation Measure: Rigorous inspections will be conducted on all rocket components to ensure structural integrity before and after assembly. Proper handling protocols will be strictly enforced to prevent disassembly or damage during transportation and pre-launch preparations.

Launch Phase:

- **Ignition Malfunction**

Mitigation Measure: Pre-flight checks and testing will be conducted to verify the functionality of ignition

- **Range Safety Issues**

Mitigation Measure: Strict adherence to range safety protocols will be enforced to maintain proper clearance, minimum of 9.1 meters from the Launch Pad, this distance is stated in America's National Association of Rocketry for type F Rockets.

- **Launch Pad Malfunction and Objects Hitting Participants**

Mitigation Measure Inspections of launch pad infrastructure will be conducted to identify and address potential malfunctions or hazards

Unfortunately, there are no specific mitigation measures for post-launch risks. We rely on our pre-launch preparations and the aerodynamic design of the rocket. We must hope that the aerodynamics are favorable and that our systems do not fail during flight.

This safety plan outlines the essential measures and considerations to ensure a successful and safe launch of our model rocket. While we can mitigate many risks during the pre-launch and launch phases, some uncertainties in the post-launch phase require us to trust in our design and preparation.

5. Materials and Cost Balance

At the start of the project the team decided on a maximum budget of 60 euros, meaning each member would have to contribute 10 euros. We felt that this was a reasonable and affordable amount for everyone, but it would also require effort on our side to make sure we didn't exceed it.

Every member agreed pretty quickly that we needed to manufacture all our components and try to use/reuse anything that we could find at our homes to build such components. Not only was this the best approach to meet our financial goals it was also a valuable experience for everyone to develop skills that could be used in future ventures.

Comparison - Mid-power

Product	P/N	Weight	Material	Shape	Desc. Rate	Shroud Line Len.	Canopy Area	Price	Buy Now!
		Sele	Selec	Selei	Sele	Select	Select	Sele	
24" Printed Nylon Parachute	29093	--	--	--	--	--	--	$10.28	Add to Cart
24"/32" Cut-to-Size Plastic Parachute	29101	0.388oz (11g)	Plastic	Octagon	varies	32" (81.28cm)	5.891ft^2 (848.3in^2) (0.543m^2)	$7.75	Add to Cart
30" Printed Nylon Parachute	29094	--	--	--	--	--	--	$15.50	Add to Cart
36" Printed Nylon Parachute	29095	--	--	--	--	--	--	$18.69	Add to Cart
48" Printed Nylon Parachute	29096	--	--	--	--	--	--	$25.73	Add to Cart
58" Printed Nylon Parachute	29097	--	--	--	--	--	--	$33.08	Add to Cart

Figure 29 – Examples of prices of parachutes in online stores

In the picture above we can see that had we decided to buy the parachute, that alone could eat up to 1/2 the project's budget (Note: the prices above don't include shipping cost).

The following table there is a list of the acquired materials during the project for GENESIS construction, indicating their description, from where they were bought, the price, the quantity purchased and the total amount (in euros) including tax and shipping, if bought online.

Item - Description	Purpose	Vendor Name	Base Unit Price	Quantity	Total (Including Tax and Shipping)
Nylon	Parachute	Casa dos Tecidos (Guimarães	6€	1	6 €
3D printing - Nose Cone, Payload and Screws		UM Library	18 €	1, 1, 4, respectively	18 €

Balsa Board (4*100*1000)	Fin Construction	Ponto das Artes (Porto)	3.55 €	1	3.55 €
Cotton thread	Shock cord	Leroy Merlin	2.68€	1	2.68 €
Mars chocolate bar	Payload	Continente	0.75€	1	0.75 €
Sticker	Decoration	UM Print shop	0.5 €	1	0.5 €
Spray Paint	Fin coating	Leroy Merlin	2€	1	2 €

Table 2 – Table of costs

It is important to note that the table in the CDR report differed from the one presented here. Some costs included in the CDR are absent from this table because we returned certain items. We made these returns because some items would not arrive in time for the rocket's construction, and we found more affordable alternatives for others.

If we add everything up it brings the total cost 33.48 €, this means we only spent roughly 55.8% of our budget. We are extremely satisfied with this figure since we even managed to include cosmetic related things like the spray paint without hurting the budget. Also is important to mention that the cost containment was achieved by pooling materials that each member of the group was willing to provide, such as materials and tools to help with assembly.

6. Final Model Design

Figure 30 - Image of the final model of GENESIS

The following table provides a detailed breakdown of the weight of each component of the rocket. All weights are measured in grams (g). This information is crucial for ensuring the proper balance and performance of the rocket, as well as for calculating the total mass, which is 483 g.

Component	Weight
Nose Cone	37 g
Payload Bay and Screws	22 g
Body Tube	175 g
Inner Tube	23 g
Motor Mount	36 g
Motor Block	2 g
Centering Ring 1	0.5 g
Centering Ring 2	0.5 g
Fins (4)	7 g
Parachute	59 g
Shock Cord	1 g
Wadding	7 g
Mars Bar	48 g
Added weight	75 g
TOTAL 483 g	

Table 3 - Components and their respective weight

After finishing our build, we went back to OpenRocket and updated every component, overriding the predicted masses with our measurements.

The updates did not affect the position of the centre of pressure, as it is solely based on the rocket's geometry, which wasn't changed since our critical design. The centre of gravity position, however, did change its position, and the total mass of the rocket changed too.

These updated values, along with new simulations, were used to determine how much mass we needed to add to the nose cone, which turned out to be 75 g.

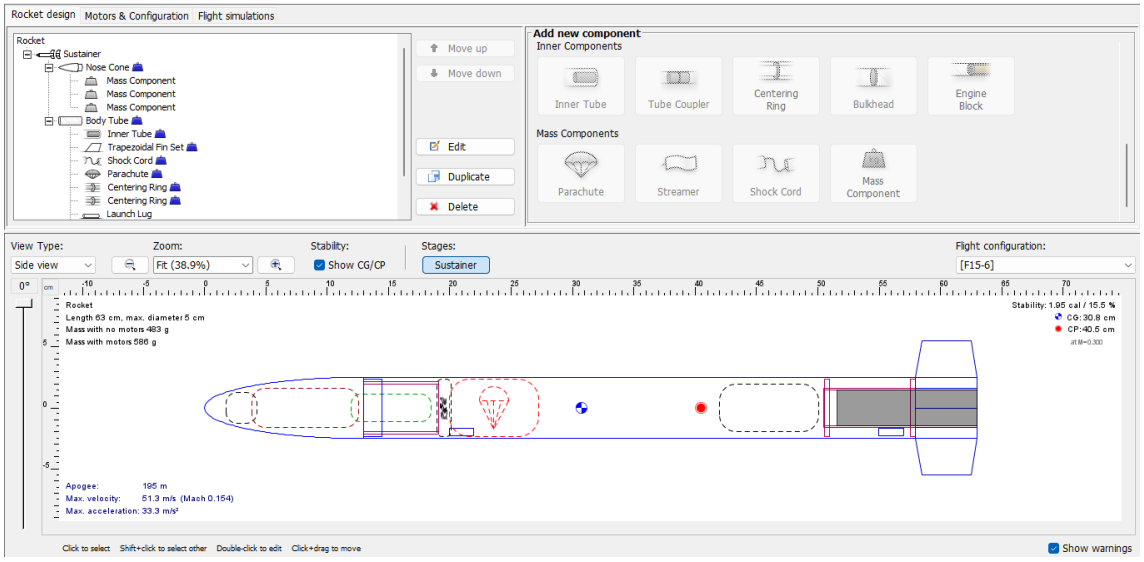

Figure 31 – Design of GENESIS with the definitive characteristics by OpenRocket

For this final configuration, we reran flight simulations for different wind conditions, assuming a rod length of 1 m (which was indicated to us by Professor Gustavo).

Windspeed (m/s)	Apogee (m)	Velocity off rod (m/s)	Ground hit velocity (m/s)
0	208	8.62	4.79
2	199	8.62	4.73
4	180	8.62	4.68
6	153	8.62	4.76

Table 4 – Results of the simulations for different wind conditions

A few things to note about our results:

- Even though the expected apogee height for a windspeed of 0 m/s is above 200 m, this is not of concern. These wind conditions are rarely observed in Cerval, and even if they occur, we can easily add slightly more payload weight on the sight.

- Even though the results for 6 m/s wind seem to meet our requirements, we wouldn't be fully confident launching in these conditions. This is because the predicted altitude is too close to our lower bound, and some of our surfaces might be a little rougher (and thus more skin friction drag inducing) than what the software anticipates. A higher windspeed also increases our concerns regarding off rod stability. As we've discussed in our critical design review, our rocket's relatively large mass will result in a low off rod velocity, meaning less stability at the start of the flight. Since coming to this conclusion, we've made great efforts to mitigate the risks, but the best preventative measure would be not to launch in adverse wind conditions. With that being said, we do think that even at these conditions we would have a very high probability of meeting our project requirements.
- At nominal conditions, we are fully confident that our rocket will perform excellently.

7. SWOT Analysis

Strengths	Weaknesses
• Diversity of people and complementary skills • Solid and structured theorical research • Preference for sustainable materials • Low project cost	• Limited resources and budget restricting the scope and complexity of rocket design • Lack of guidance and supervision during work that should be safety-conscious • Difficulty juggling schedules
Opportunities	**Threats**
• Application of the knowledge acquired in the various subjects • Establishing relationships with different partner organisations • Cooperation and collaboration	• Restrictions imposed • Lack of proper place to manufacture the rocket • Difficulties in transporting and storing the rocket

Table 5 – SWOT Analysis

8. Conclusion

Several months have passed since the teachers set us the challenge of building a model rocket, and we have an undeniably positive assessment of our journey, although it was far from straightforward.

Right from the start, we faced numerous challenges, which quickly fostered a strong team spirit among us. This camaraderie, coupled with the expertise we gained from professionals and extensive research, allowed us to overcome the obstacles we encountered. We are now more adept at selecting the most relevant information for scientific research, posing insightful questions to field experts, meticulously analysing data, swiftly identifying flaws in our prototype, collaborating more effectively as a team while respecting individual differences, and communicating our findings to others.

Stepping out of our comfort zones, we displayed resilience in the face of various difficulties and concluded the project feeling proud and assured that we had given our best effort.

In essence, navigating these uncharted waters allowed us to successfully embark on our exploration of the aerospace field. This experience was filled with thrilling moments and, most importantly, has inspired us to continue unveiling the marvels of aerospace science.

Finally, we extend our heartfelt gratitude to the teachers who supported us throughout this endeavour. We also thank our peers for their collaborative spirit. Despite the competitive nature of building the best rocket, they were always ready to lend a helping hand when needed.

Figure 32 - Picture of the MARS team with GENESIS

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